TalentVerse

Skills & Career Intelligence · CA3 Generative AI Foundations · Option 9

1. Problem statement

Students and early-career candidates hold their career information in scattered, unstructured forms: a resume, a spoken introduction, a job posting they are eyeing, and rough notes from interview practice. Turning this into action is slow and guess-driven. They struggle to see their own skills clearly, to know what a target role actually requires, to find out which gaps matter most, to choose a project that proves their ability, and to learn from interview practice in a structured way. Most tools help with only one of these steps, and none connects them around a single, reviewable candidate profile.

2. Our approach

TalentVerse converts messy input into structured data first, then builds every other feature on that profile. Each AI call is constrained by a response schema, checked by server-side validation, and saved to a database before anything is shown to the user.

- **Profile extraction:** resume text or an audio introduction becomes a structured profile (skills, education, experience, projects, certifications) plus a list of missing information.
- **Skill-gap map and learning plan:** a career goal is compared against the profile and turned into prioritised gaps and sequential milestones with success criteria.
- **Job match:** role requirements are extracted from a job description and compared with the profile, with evidence quoted for every matched skill.
- **Portfolio brief:** interests and existing skills produce an achievable project brief that stretches the candidate in a few new areas.
- **Interview feedback:** practice notes become strengths, improvement areas and small next-practice tasks.

Stack: HTML/CSS/JavaScript frontend, Python Flask REST API, Gemini API with JSON response schemas, server-side validation, SQLite storage, deployable on Render. Prompts instruct the model to use only supplied facts and to treat user text as data, not instructions.

3. Why TalentVerse, and why other tools can't do the same

Other tools are good at their own slice. The gap is that none of them chains all five steps through one structured, evidence-based profile.

Alternative	What it does well	What it does not do
Generic chatbots (ChatGPT, Gemini chat)	Flexible answers to any prompt.	No fixed schema, no saved profile, nothing carried between steps. Each answer is free text the user must re-paste and re-check, and results are not stored or comparable.
Resume parsers / ATS software	Extract fields from resumes for recruiters.	Built for employers, not candidates. They stop at extraction: no gap analysis, learning plan, project idea or interview coaching.
Job platforms (LinkedIn, Indeed)	Large job listings and keyword matching.	Keyword matching only. No evidence-backed explanation of why a skill matches or what is missing, and no plan to close the gap.

Alternative	What it does well	What it does not do
Course platforms (Coursera, Udemy)	Teach skills.	Do not read the candidate's actual profile or the specific job, so recommendations are generic.
Interview practice apps	Simulate interviews.	Rarely connected to the candidate's profile or target role, and do not turn the user's own notes into next steps.

What makes TalentVerse different

- **One profile, five outcomes:** every feature reuses the same validated profile instead of starting over.
- **Structured and validated:** schema-guided output is rejected, not saved, if fields are missing or wrongly typed.
- **Evidence over claims:** matched skills quote the profile; the model must not invent employers, dates or skills.
- **Stored and reviewable:** profiles and analyses persist in the database, so results can be revisited and compared.
- **Candidate-first:** built to help the individual improve, not to rank or screen them.

Honest limitations

Other tools could add these features, and a determined user can approximate this workflow by hand with a chatbot. TalentVerse's value is combining the steps into one repeatable, validated workflow. AI output can be incomplete or wrong, so users should review it. Match percentages and skill levels are guidance, not hiring decisions. Resume content is sent to an external AI service, so demos should use sample data. The app has no user authentication yet.